IIIT BHUBANESWAR
Imagine, Innovate, Inspire

Asish Kumar Patra



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OBJECTIVE

Aspiring Software Development Engineer pursuing dual degrees at IIT Guwahati and IIIT Bhubaneswar with hands-on experience in building scalable software systems, full-stack applications, and data-driven solutions. Equipped with strong foundations in Data Structures & Algorithms, object-oriented programming, and problem solving, I seek to contribute to the design and development of high-quality, efficient, and impactful software products while continuously expanding my engineering expertise.

PROFESSIONAL SYNOPSIS

- Dual-degree student at IIT Guwahati and IIIT Bhubaneswar specializing in Data Science, Artificial Intelligence, and Cloud Computing.
- International research and industry experience across McLaren Racing, IST Austria, Stanford University, Microsoft Elevate, and IIT Guwahati.
- Skilled in Machine Learning, Data Analytics, Computer Vision, NLP, Cloud Computing, and Full-Stack Development.
- Strong competitive programming background with 1900+ LeetCode problems solved and Top 0.1% global ranking (as per LeetCode Rewind 2026).
- Passionate about building AI-driven systems for healthcare, motorsports, finance, and intelligent automation.

EDUCATIONAL QUALIFICATIONS

Qualification	School/College	Board/ University	Year of Passing	Percentage/ CGPA
B. Tech ETC	IIIT Bhubaneswar	IIIT Bhubaneswar	2028	7.81
B. Sc (Hons.) Data Science and AI	IIT Guwahati	IIT Guwahati	2027	8.25
Class XII	DAV Public School, Unit-8, Bhubaneswar	CBSE, New Delhi	2023	91.2%
Class X	DAV Public School, Unit-8, Bhubaneswar	CBSE, New Delhi	2021	94.2

WORK EXPERIENCE

- **McLaren Racing, Woking, UK**
  Student Racing Data Scientist Intern
  May,2026 – July,2026
  - Analyzed high-frequency Formula 1 telemetry data using Python, SQL, and statistical methods to evaluate driver performance, vehicle dynamics, and race strategy
  - Developed real-time telemetry analysis pipelines and interactive dashboards for visualizing throttle, braking, steering, tyre degradation, and lap performance
  - Applied machine learning, time-series analysis, and predictive modelling to identify performance trends and support data-driven engineering decisions.
  - Collaborated with race engineers and data scientists on telemetry interpretation, aerodynamic correlation, and race simulation workflows to optimize on-track performance.
  - Designed and optimized software components for telemetry data processing by implementing efficient algorithms, reusable modules, and maintainable code structures to support high-performance analytical workflows.
  - Authored technical documentation and collaborated in code reviews to ensure software quality, maintainability, and seamless integration of analytics solutions within cross-functional engineering teams.

- **IIT Guwahati, Guwahati, India**
  Project Intern
  March 2026 – July 2026
    - Developed AI and machine learning solutions by designing end-to-end data processing pipelines, performing feature engineering, and optimizing predictive models using Python and Scikit-learn.
    - Conducted applied research through data analysis, model evaluation, and technical documentation, delivering scalable solutions for real-world analytical and research applications.
- **Microsoft Elevate/AICTE**
  Power-BI Trainee
  February 2026 – March 2026
    - Developed an interactive **Air Quality Monitoring Dashboard** in **Power BI** by analysing AQI trends across Indian cities using Power Query, DAX, and data visualization techniques.
    - Designed dynamic dashboards featuring KPI cards, trend analysis, pollution maps, city rankings, and interactive filters to identify pollution patterns and support data-driven insights.
    - Performed data cleaning, transformation, and exploratory data analysis to convert raw environmental datasets into actionable business intelligence reports and visualizations.
- **Institute of Science and Technology Austria, Klosterneuburg, Austria**
  Research Intern ISTernship 2025
  May 2025 – June 2025
    - Conducted research on **Optimal Code Block Design** under Prof. Marco Mondelli, investigating compiler co-optimization strategies that balance modular software design with runtime performance.
    - Collaborated with researchers at **ISTA** and engineers at **Google Zurich** as part of an international exchange, contributing to compiler optimization research through formal methods and architecture-aware system design.
    - Developed and evaluated prototype optimization workflows using **C++, Python, LLVM, Graphviz, Git, and performance profiling**, analysing the impact of code organization on compiler behaviour.
    - Performed experimental analysis of compiler diagnostics and optimization trade-offs, validating approaches for improving execution efficiency across complex software systems.
    - Documented research findings through technical reports and presentations, delivering scalable compiler optimization methodologies with applications in high-performance computing and modern software engineering.
- **Stanford University**
  Code in Place 2025 Section Leader
  April 2025 – June 2025
    - Selected as a **Section Leader** from a global applicant pool to mentor **100+ students** in Python programming through Stanford's **CS106A** curriculum, providing personalized guidance, debugging support, and interactive learning sessions.
    - Built **Dream Decoder**, a Python-based NLP application that analyses dream narratives using sentiment analysis and symbolic interpretation, reinforcing core programming concepts while mentoring first-time programmers.
- **Outlier.AI**
  Generative AI Engineer
  February 2025 – June 2025
    - Developed and evaluated **Generative AI** workflows through prompt engineering, response validation, and large language model optimization to improve output quality and reliability.
    - Engineered AI-powered solutions by integrating NLP techniques, cloud-based workflows, and model evaluation frameworks to support scalable real-world applications.
    - Collaborated with cross-functional teams to fine-tune foundation models, analyse model performance, and maintain high-quality AI-generated outputs across diverse use cases.

PROJECTS

- **F1 Telemetry Analytics Platform**
  GitHub Link: ultron513/F1-Telemetry-Dashboard-2025
    - Developed a full-stack Formula 1 telemetry analytics platform for professional race performance analysis, enabling interactive visualization of driver telemetry, lap comparisons, race strategy evaluation, and predictive insights using real-world F1 data.
    - Tech Stack: Python, FastF1, Fast API, PostgreSQL, SQL Alchemy, React.js, Next.js, TypeScript, Recharts, REST APIs, Docker, Git.
- **Natural Sound Intelligence Platform**
  GitHub Link: ultron513/Natural-Sound-Intelligence-Platform
    - Built an AI-powered environmental sound classification platform for automated recognition of natural sounds, featuring real-time prediction, interactive analytics dashboards, and comprehensive dataset visualization for research and intelligent monitoring applications.

- Tech Stack: Python, Scikit-learn, Librosa, NumPy, Pandas, Fast API, Streamlit, Matplotlib, Plotly, Joblib, HTML, CSS, Git.

- **Credit Card Fraud Detection & Financial Risk Intelligence Platform**
  GitHub Link: ultron513/Credit-Card-Fraud-Detection-Financial-Risk-Intelligence
  - Designed an end-to-end financial fraud detection system that leverages machine learning and anomaly detection techniques to identify fraudulent transactions, assess financial risk, and support secure decision-making for banking and fintech applications.
  - Tech Stack: Python, Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn, SHAP, SMOTE, Jupyter Notebook, Git.
- **Road Damage Detection & Infrastructure Assessment System**
  GitHub Link: ultron513/Road-Damage-Detection-Infrastructure-Risk-Intelligence
  - Developed a computer vision-based infrastructure assessment system capable of detecting and classifying road surface damage from images to support automated road inspection and smart city maintenance initiatives.
  - Tech Stack: Python, TensorFlow, Keras, OpenCV, Albumentations, Grad-CAM, CNNs, Transfer Learning, NumPy, Matplotlib, Flask, Git.
- **Air Quality Monitoring Dashboard**
  GitHub Link: ultron513/AQI-PowerBI-Dashboard
  - Created an interactive business intelligence dashboard for monitoring air quality trends across Indian cities, enabling pollution analysis, environmental reporting, and data-driven policy insights through dynamic visualizations.
  - Tech Stack: Microsoft Power BI, Power Query, DAX, Microsoft Excel, SQL, Data Modelling, Data Visualization.
- **Dream Decoder**
  GitHub Link: ultron513/dream-canvas-interpret
  - Built an NLP-powered application that interprets user dream narratives by analyzing emotional tone and symbolic patterns, providing intuitive textual and graphical insights through a rule-based interpretation engine.
  - Tech Stack: Python, NLTK, TextBlob, Tkinter, Turtle Graphics, Rule-Based NLP, Object-Oriented Programming, Git.

ACADEMIC ACHIEVEMENTS

- Ranked among the **Top 0.1% programmers globally on LeetCode**, solving **1800+ algorithmic and data structure problems** across diverse domains.
- Achieved a **Specialist rating on CODEFORCES**, demonstrating consistent performance in competitive programming contests.
- Pursuing dual undergraduate degrees in **Data Science & Artificial Intelligence (IIT Guwahati)** and **Electronics & Telecommunication Engineering (IIIT Bhubaneswar).**
- Earned industry-recognized certifications including **Google AI Essentials, Google Cybersecurity, Google IT Automation with Python, AWS S3 Basics,** and **GIS Mapping & Spatial Analysis**.
- Solved algorithmic challenges spanning Dynamic Programming, Graphs, Trees, Greedy Algorithms, Binary Search, and Advanced Data Structures, strengthening problem-solving proficiency for software engineering.

COMPUTER PROFICIENCY

- **Programming Languages:** Python, Java, C++, C, JavaScript, TypeScript, SQL
- **Software Development:** Data Structures & Algorithms, Object-Oriented Programming (OOP), React.js, Next.js, Fast API, Flask, HTML5, CSS3, REST APIs, Streamlit
- **Databases, Cloud & Developer Tools:** PostgreSQL, MySQL, MongoDB, Google Cloud Platform (GCP), Docker, Git, GitHub, Linux, VS Code, Jupyter Notebook
- **AI, Machine Learning & Data Analytics:** Scikit-learn, TensorFlow, Keras, OpenCV, XGBoost, Librosa, NumPy, Pandas, Matplotlib, Plotly, Power BI, Power Query, DAX

INTERESTS / EXTRACURRICULAR ACTIVITIES

- Mentor and collaborative team player with experience guiding peers, conducting technical discussions, and contributing to knowledge-sharing initiatives.
- Passionate about open-source technologies, continuous learning, and exploring emerging trends in software engineering and artificial intelligence.
- Regular participant in university technical festivals, coding workshops, and innovation-focused events.
- Enthusiastic **Formula 1** follower with a keen interest in motorsports, race strategy, and performance engineering.
- Avid reader of technology, software architecture, and artificial intelligence research articles.